Lecture plan · for the lecturer

Evaluating models and assessing students

AI · University · Lecture 4

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Evaluate without leakage.
- Report beyond accuracy.
- Assess learning when LLMs are allowed.

Activity: Audit a project

1. Warm-up (5 min): Show a leaderboard where the winner tuned on test and ask students what the score means.
2. Teach (10 min): Show the three splits and a confusion matrix on imbalanced data.
3. Audit (20 min): Teams audit a sample project for leakage, metric choice and subgroup gaps.
4. Redesign (20 min): Rewrite one assignment so a language model cannot complete it alone.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- What leak did you find?
- Which metric would you require?
- What is your policy on LLM use?

Exit ticket

- Write one oral-defence question for your next ML project.

Try it in your next lecture
Add a ten-minute oral defence to your next project: each student explains one error their model makes and one design choice they would change.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Fraud is 1 in 1,000. A model flags nothing and has 99.9% accuracy. Which metric exposes it?**

◯ Training loss ◯ Accuracy ◯ Recall

2 **Work out the metrics. A model makes 40 positive predictions; 30 are right. There were 60 real positives.**

Precision = _____ %. Recall = _____ %.

Word bank: 60 · 50 · 75 · 30

3 **Which assessment designs still measure a student's own understanding when language models are allowed? Choose all that apply.**

☐ A graded experiment log ☐ Error analysis on data they collected

☐ A take-home task to code a standard classifier ☐ An oral defence of their notebook

4 **A model card should report metrics broken down by relevant subgroups.**

◯ True ◯ False

5 **Match each project flaw to the evidence that would reveal it.**

Tuned on the test set	• • • •	Near-identical rows found in train and test
Duplicate records across splits	• • • •	Score drops on a fresh held-out set
Fails a subgroup	• • • •	An oral defence
Student can't explain the code	• • • •	Metrics disaggregated by group

Answers · for the lecturer

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1 **Recall**

Recall is 0 because it catches none of the fraud.

2 **Precision = 75 %. Recall = 50 %.**

Precision = 30/40 = 75%; recall = 30/60 = 50%.

3 **An oral defence of their notebook; Error analysis on data they collected; A graded experiment log**

A standard take-home coding task can be completed by a tool; the others need the student's own reasoning about their own work.

4 **True**

Disaggregated evaluation is central to model cards because averages can hide large gaps.

5 **Tuned on the test set — Score drops on a fresh held-out set**
Duplicate records across splits — Near-identical rows found in train and test
Fails a subgroup — Metrics disaggregated by group
Student can't explain the code — An oral defence

Tuning on the test set inflates its score, so only a fresh held-out set shows the drop; duplicates leak answers across splits, so look for near-identical rows; an average hides a failing subgroup until metrics are split by group; and only an oral defence shows whether students understand their own code.

For you to think about

Which of your current ML assessments could a language model complete? What would you change first?